Functional Description of Diagnosis

ZOTYE A01

ABS Gen9

Protocol: KWP2000 On CAN

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1 History

All corrections of misspelling are excluded of the following list.

Ver.	Date	Editor	Chapter	Description of changes
01.00	18.10.2010	Sathish Muthu S	All	Initial Version Based on China mainstream DIAF v1.1
01.01	15.03.2011	Sathish Muthu S	4 5.5.2 5.7.2 6 6.1.1 6.2 6.3.4 6.4.1 6.4.1 & 6.5.1 6.5.3 6.6.3 6.7.1 6.9.2 6.10.1 6.10.2 6.10.6 6.10.6 & 6.10.7 6.10.12.1 6.10.12.2	- Abbreviation table is added. - P2* time is updated to 5000ms and P3max time is updated to 10000ms. - Stop Diagnostic session is removed and ECU Reset is added. - P3max is updated as P3 CAN-ECU (MAX). - P3 CAN-ECU (MAX) value is updated to 10 seconds. - Service sections are updated. - Service Stop Diagnostic session (SID 20h) is removed and ECU reset (SID 11h) is added. - In service Start routine by LID, Periodic Actuator control (LID 02) is split in to Evac and Fill control (LID 02h) and Repair Bleed control (LID 03h). - LIDs for Actuator test, Dynamic test and Wheel speed sensor test are updated. - "StandardDiagnosticMode" is updated as "DefaultDiagnosticMode". - ECU Reset service description is added. Stop Diagnostic session service description is removed. - P2max is updated to P2 CAN-ECU. - Description for "NumberOfFaultDetection" is modified. - Comment for bit4 is updated. - "DTCWarningLampCalibrationStatus" is changed to "DTCWarningLampStatus". - In third remark byte2 is updated to byte1. - Vehicle name and Customer part number is updated. - BB number, version numberand supplier part number data type representation has been added. - Byte(size) is modified as Size(Byte). - Bit position is updated in Remark. Bit32 -> Bit31 and Bit31 -> Bit30. - Remark 2 related to wheel and vehicle speed is added. - Remark 1 is updated. - In Remark 2, control status is updated to Control in progress(03h). - Supported LID table is updated. - Remark 1 system 8 is updated to system9. - Description for Periodic actuator control is removed. - Description for Evac and Fill and Repair Bleed is added. - Control status 03h and 05h are added. - Bit position is updated in Remark. Bit32 -> Bit31 and Bit31 -> Bit30. Supported data table

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				is updated.
			6.10.12.3	- LID 03h is added.
			6.10.12.5 & 6.10.12.6	- Remark related to the byte 3...10 is added.
			6.10.12.7	- Value of speed returned when WSS faulure detected is updated from FFFFh to 7FFFh.
			6.7.4 & 6.11.4	- Remark for NRC 78h is added.
			7.1	- DTC 0x501008 - ECU error (general software problem) is added.
			8	- Diagnostic flow chart is updated with supported services.
				The above changes are done based on the latest China MS Gen9 KWP on CAN DIAF v1.8

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2.1 International and National Documents

The general requirements of CAN diagnosis communication protocol are based on the Keyword Protocol 2000 Design Standard requirements, specified in the following documents:

1. ISO 14229 : Road Vehicles – Diagnostic Systems: Diagnostic service specification
2. ISO 14230-1: Road Vehicles – Diagnostic Systems: KWP2000 – Physical layer
3. ISO 14230-2: Road Vehicles – Diagnostic Systems: KWP2000 – Data Link Layer
4. ISO 14230-3: Road Vehicles – Diagnostic Systems: KWP2000 - Part 3 Implementation
5. ISO 15765-2: Road Vehicles – Diagnostics on CAN: Network Layer Services
6. ISO 15765-3: Road Vehicles – Diagnostics on CAN: Application Layer Services

Remark:
Requirements in the above referenced documents may not be fully implemented. The document Functional Description of Diagnosis is the agreed specification for the implemented customer diagnosis.

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3 Introduction

CAN diagnosis provides the capability of quickly accessing the information and functions of the control unit over the **C**ontroller **A**rea **N**etwork (CAN), without disconnecting the cable harness. CAN diagnosis provides services to:

- Identify the ECU
- Read input signals of the ECU
- Write output signals to the actuators
- Read out ECU-internal information, such as the contents of the fault memory

CAN diagnosis can be used for end-of-line testing or in the servicing workshop. The testing time can be reduced substantially by using the services provided by CAN diagnosis.

This document describes the diagnosis functions and specifications of the Normal, EOL and project specific Diagnostic Session (Application Software).

4 Abbreviations

Abbreviations	Description
ABS	Antilock Braking Systems
AV	Outlet Valve
BLS	Brake Light Switch
DTC	Diagnostic Trouble Code
EBD	Electronic Brake force Distribution
ECU	Electronic Control Unit
EV	Inlet Valve
IGN	Ignition
LSB	Least Significant Bit
MSB	Most Significant Bit
PID	Parameter Identifier

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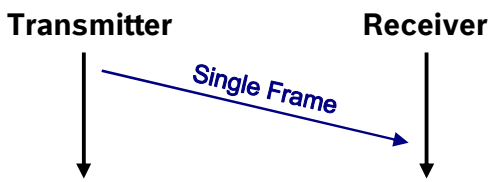
5 Communication on CAN

5.1 Data Stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly the following two types of CAN data streams are supported:

5.1.1 Non-segmented data stream

Consists of a Single Frame (SF) of non-segmented CAN message. This can be used for messages with maximum data length of 7 data bytes



5.1.2 Segmented data stream

A segmented data stream consists of multiple frames of segmented CAN messages. A segmented data stream must be used if the maximum data length exceeds 7 data bytes in case of normal addressing.

The data stream comprises a First Frame (FF) and a number of Consecutive Frames (CF) containing the subsequent data bytes, following the First Frame. The receiver has to send a Flow Control (FC) after reception of the First Frame.

The Flow Control contains information regarding Block Size and the Separation Time. The Block Size (BS) is the maximum number of Consecutive Frames the receiver allows the sender to send, before waiting for an authorisation (Additional Flow Control Frames) from the receiver to continue transmission. The Separation Time (Stmin) is the (minimum) time the sender has to wait between the transmissions of two consecutive frames.

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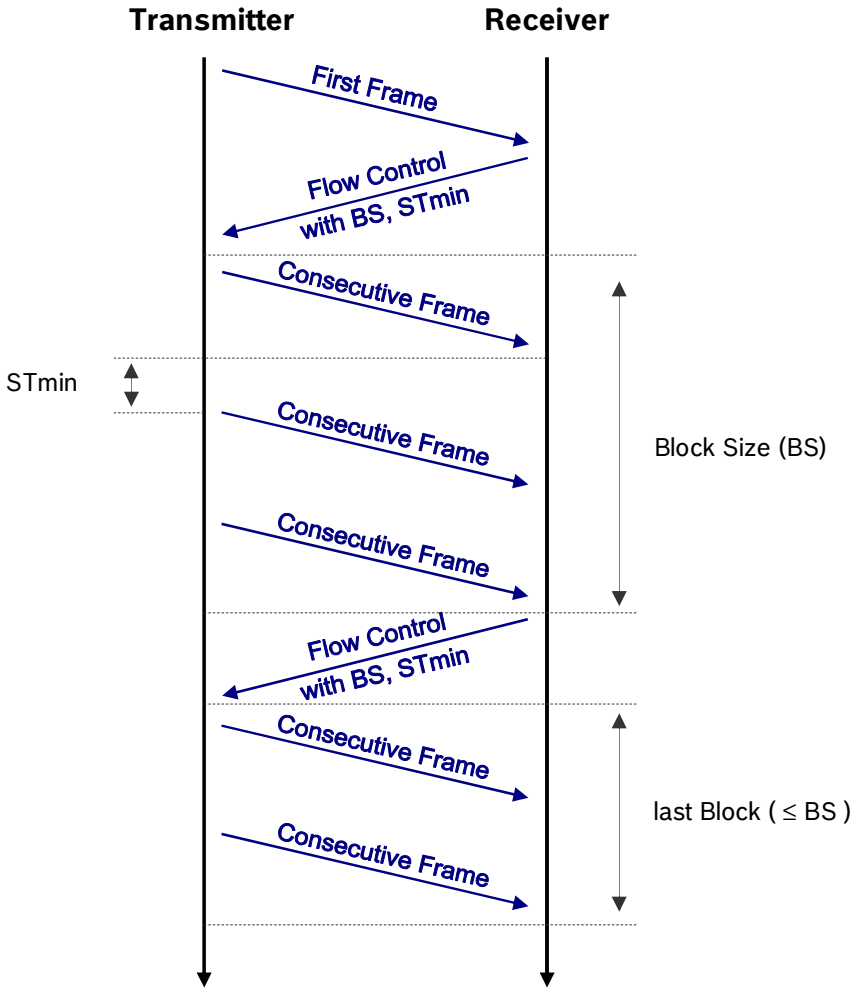

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5.2 Message Types

Each Single Frame (SF), First Frame (FF), Consecutive Frame (CF) or Flow Control (FC) has an 8-byte Data Field (DLC of CAN message is set to 8) consisting of the following parts:

- Protocol Control Information (PCI)
  indicates the frame type, number of frames, data length etc.
- Transmission data (Data)
  contains data to be sent

Data Field Structure					
Frame Type	**Byte 1**		**Byte 2**	**Byte 3**	**Byte 4 – 8**
	Bits 7-4	**Bits 3-0**			
Single Frame (SF)	0000	SF_DL	DB 1	DB 2	DB 3 – 7
First Frame (FF)	0001	FF_DL		DB 1	DB 2 – 6
Consecutive Frame (CF)	0010	SN	DB x_1	DB x_2	DB x_3-x_7
Flow Control (FC)	0011	FS	BS	Stmin	N/A

SF_DL: 4 Bit Single Frame Data length
SN: Sequence Number
BS: Block Size
DB: Data byte
FF_DL: 12 Bit First Frame Data length
FS: Status of data stream
Stmin: Minimum Separation Time

5.2.1 Single Frame

Single Frame message type is used for data blocks with a maximum data length of 7 Bytes and is identified by the upper nibble of PCI set to 0000b. Single Frame Data length SF_DL is computed as:

Single Frame Data Length (SF_DL):

Hex Value	Description
0h – 7h	Single Frame Data length (SF_DL) contains number of data Bytes

5.2.2 First Frame

First Frame message type is used for data blocks more than 7 Bytes long, in combination with one or more Consecutive Frames and is identified by the upper nibble of PCI set to 0001B. First Frame Data length FF_DL is computed as:

First Frame Data Length (FF_DL):

Hex Value	Description of FF_DL
000h – 007h	Reserved
008h – FFFh	First Frame Data length (FF_DL) contains number of data bytes. 12 Bit FF_DL theoretically supports sending of up to 4096 data bytes. However the current implementation does not need to support this maximum of 4096 data bytes.

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5.2.3 Consecutive Frame

Consecutive Frames follow First Frame and is identified by the upper nibble of PCI set to 0010B. The lower nibble of the PCI has a Sequence Number (SN) which is determined as follows:

- First Frame is regarded as containing Sequence Number 0.
- Hence the first Consecutive Frame always holds Sequence Number 1.
- Sequence Number is incremented by one for every additional Consecutive Frame.
- A Flow Control frame sent "in between" does not affect the Sequence Number counting.
- When the Sequence Number reaches the value of 15, the following Consecutive Frame will be sent with Sequence Number set to 0.

Definition of Sequence Number (SN):

Protocol Data name	FF	CF	CF	CF	CF	CF	CF	CF	CF
SN (hex)	0h	1h	2h	...	Eh	Fh	0h	1h	...

Hex Value	Description
0h – Fh	Sequence Number (SN) The Sequence Number has a range of 0 to 15. SN-Value is stored to the lower 4 bits of the Byte 1.

5.2.4 Flow Control

After the First Frame or the last Consecutive Frame of a block of frames when further CFs are required to complete the data transfer, a Flow Control message must be sent by the receiver node, and is identified by the upper nibble of PCI set to 0011b. With the Flow Control, the receiving node notifies the transmitting node how many messages it can receive between two Flow Control frames (maximum Block Size, Bsmax) and at what rate between Consecutive Frames (minimum Separation Time, Stmin). The lower nibble of the Flow Control PCI shall have a Flow Control Status (FS) set to "Continue To Send (CTS)".

Stream Status (FS):

Hex Value	Description
0h	Continue To Send (CTS) The Receiver is ready and awaits the BS announced maximal amount of Consecutive Frames
1h	Wait (WT) The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "1". It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer.
2h	Overflow (OVFLW) The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "2". It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU

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	exceeds the buffer size of the receiving entity.

Block Size (Bsmax):

Hex Value	Description
00h	Block Size (Bsmax) No more Flow Controls will be sent during the segmented Message. All Consecutive Frames will be sending without a stop for an additional Flow Control to the Receiver.
01h – FFh	Block Size (Bsmax) Defines the maximal amount of Consecutive Frames till next Flow Control.

Separation Time (Stmin):

Hex Value	Description
00h – FFh	Separation Time (Stmin) The Separation Time defines the minimal allowed time (in ms) between 2 Consecutive Frames.

5.3 Addressing Modes

5.3.1 Physically addressing Mode

All Physically addressed messages will be received by the ECU on ECU_PHY_ADDR (see 5.5.1 section below for supported CAN ID). It is possible to execute a service supported in that active diagnostic session of the ECU, when the request is made using this mode. Response from the ECU is sent on TESTER_ADDR.

5.4 Diagnostic Sessions

The ECU supports the following Diagnostic Sessions:

- **Normal / Default Session**
  At ignition ON, the ECU enables the Normal diagnostic session by default. In this session both the controller functions and diagnostic services are active. But this session does not permit the execution of all supported diagnostic services of the EOL and specific Diagnostic Session (see “Availability of Services” section below for services supported during this session).

- **EOL Diagnostic Session**
  The Diagnostic session for assembly line.

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5.5 Overview

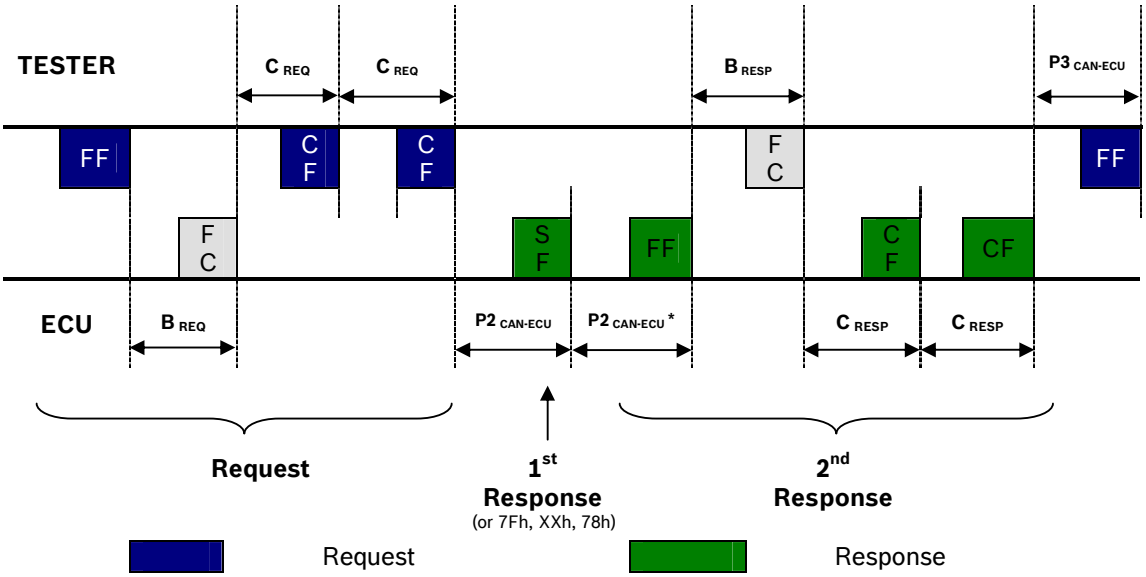
The CAN communication between Tester and ECU is only possible in one direction at a time. In a Half-Duplex diagnostic session the Tester is the master and the ECU is the slave. The master will send requests and the slave – if addressed – will answer with a response.

5.5.1 Communication Addresses

Addressing Type	Name	Hex Value ZOTYE A01
Tester CAN receive message ID for Diagnosis	TESTER_ADDR	712h
ECU CAN receive message ID for Physical Addressing	ECU_PHY_ADDR	702h

5.5.2 Communication Timing Parameters

Network Layer Timing:



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Parameters for Physical Layer, Data Link Layer and Network Layer:

Parameter	Name	Value	
Communication speed (baud rate)	BR-CAN	500 kBd ± 2 kBd	
Max Timeout to access the network (time for transmission of the CAN frame on the sender side)	As	25 ± 2 ms	
Blocksize – the maximum number of Consecutive Frames the receiver allows the sender before waiting for a Flow Control message	Bsmax	0	
Separation time – The Receiver can set the Stmin value as 10ms. The Transmitter shall wait for more than Receiver Stmin. Value before sending the next Consecutive Frame	Stmin	10ms	
Tester Request		Min	Max
Time between reception of First Frame from Tester and transmission of Flow Control from ECU. The maximum timing is defined under the condition that the CAN Bus load is able to process the ECU transmission.	B_{REQ} (Br)	NA	50 ms#
Time between transmission of Flow Control and reception of next Consecutive Frame or Time between reception of two Consecutive Frames	C_{REQ} (Cs)		50 ms#
Time between end of reception of a Tester Request and start of an ECU-Response	$P2_{CAN\text{-}ECU}$	0 ms	50 ms
Time between end of transmission of an ECU response with Negative Response Code 78h (Response Pending) and start of next ECU response	$P2_{CAN\text{-}ECU}$*	0 ms	5000 ms
ECU Response		Min	Max
Time between transmission of First Frame and reception of Flow Control from Tester	B_{RESP} (Bs)	NA	100 ms
Time between reception of Flow Control and transmission of following Consecutive Frame or Time between transmission of two Consecutive Frames	C_{RESP} (Cr)	0	100 ms
Time between end of an ECU-Response and start of a Tester-Request in Normal Diagnostic Session	$P3_{CAN\text{-}ECU\text{-}StdDiag}$	NA	Infinite
Time between end of an ECU-Response and start of a Tester-Request in Extended Diagnostic Session	$P3_{CAN\text{-}ECU\text{-}ExtDiag}$	NA	10000 ms

#The maximum timing is defined under the condition that the CAN Busload is able to process the ECU transmission.

Note:
When the tester sends a request and the tester does not expect any response (e.g. functionally addressed request or a physical addressed message with “No response required” option), the tester has to wait

$$P3_{CAN\text{-}ECU\text{-}func} = P2_{CAN\text{-}ECU\ (Max)} + P3_{CAN\text{-}ECU\ (Min)} + P_{Service_Execution_Time}$$

before sending the next request.

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5.6 Communication Failures

5.6.1 Unexpected Messages

As a general rule, the arrival of an unexpected message or a message with illegal format from any node will be ignored.

5.6.2 Missing Messages

If the ECU senses a communication failure (e.g. timing violations of the Network Layer) during a tester request, the ECU discards the request and no response will be sent.

The tester has to monitor that no ECU-response has been sent within **P2** $_{\text{CAN-ECU}}$, and hence the tester can repeat his previous request.

During Extended Diagnostic Session, if no request is received from the tester for **P3** $_{\text{CAN-ECU (MAX)}}$ (criteria for broken communication wire) ECU will exit to Normal Session.

5.7 Condition and Behaviour of the ECU

5.7.1 Speed Limit

The ECU monitors a speed limit during the entire EOL diagnosis mode. Speed limit for the tester address (see 5.5.1) is 10 Km/h.

5.7.2 Exit Conditions

The ECU will terminate an active EOL diagnostic mode in the following cases:

- The speed limit is reached. Note that speed limit can be disabled by using the InputOutputControlByLocalIdentifier service (see 6.9).
- An ECU Reset request is received (See 6.2).
- The ECU does not receive any tester message for more than 10 seconds (**P3** $_{\text{CAN-ECU (MAX)}}$ Timeout)

EOL diagnostic session is started using the StartDiagnosticSession service. When an EOL diagnostic session is aborted, all EOL diagnostic actuations (eg: lamps, Valves) will be switched off.

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6.1 StartDiagnosticSession

6.1.1 Purpose

The service StartDiagnosticSession is used to select a specific ECU DiagnosticMode (called session). Depending on the session selected, actions are authorized, prohibited or limited. The following sessions are defined:

PARAMETER: DiagnosticMode		
Size (Byte)	**Description**	**Value**
1	DefaultDiagnosticMode	81h
1	ECUEOLMode	83h

Remark: In the Default Session the ABS functionality is fully available.

Remark: In the EOL mode the ABS functionality is not available. Switching to EOL mode is only possible when the ECU is not in regulation.

6.1.2 Request

TESTER → ECU: [SERVICE] - Request		
Byte	**Contents**	**Value**
0	Service Identifier	10h
1	DiagnosticMode	xxh

6.1.3 Positive Response

ECU → TESTER: [SERVICE] – Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	50h
1	Diagnostic Mode	xxh

6.1.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	10h
2	**Response Code** Sub Function Not Supported / Invalid Format ConditionsNotCorrect/RequestSeqError BusyRepeatRequest	 12h 22h 21h

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6.2 ECUReset

6.2.1 Purpose

The service ECUReset is used to restart the ECU.

PARAMETER: ResetMode		
Size (Byte)	Description	Value
1	PowerOn	01h

6.2.2 Request

TESTER → ECU: [SERVICE] - Request		
Byte	Contents	Value
0	Service Identifier	11h
1	ResetMode	01h

6.2.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	Contents	Value
0	Service Positive Response Identifier	51h
1	ResetMode	01h

6.2.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	Contents	Value
0	Negative Response Identifier	7Fh
1	Request Service Identifier	11h
2	**Response Code** SubFunctionNotSupported/InvalidFormat RequestCorrectlyReceived/ResponsePending	 12h 78h

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6.3 ClearDiagnosticInformation

6.3.1 Purpose

The service ClearDiagnosticInformation is used to clear the fault memory. The parameter GroupOfDTC is used to indicate which ECU has to clear its fault memory.

6.3.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	14h
1	GroupOfDTC – High byte	FFh
2	GroupOfDTC –Low byte	00h

6.3.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	54
1	GroupOfDTC – High byte	FFh
2	GroupOfDTC –Low byte	00h

6.3.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	14h
2	**Response Code** SubFunctionNotSupported/InvalidFormat BusyRepeatRequest RequestCorrectlyReceived/ResponsePending	 12h 21h 78h

Remark: If the erasing of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has been finished.

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6.4 ReadStatusOfDTC

6.4.1 Purpose

The service ReadStatusOfDTC is used to read the status of a specific DTC from the failure memory.

PARAMETER: StatusOfDTC		
Size (Byte)	**Description**	**Value**
1	StatusOfDTC	xxh

StatusOfDTC	
Position (Bit)	**Description**
0..3	• DTCFaultSymptom • 0000b = “No fault symptom available for this DTC” • 0001b = “Above maximum threshold” • 0010b = “Below minimum threshold” • 0100b = “No signal” • 1000b = “Invalid signal”
4	• 0b = Test completed / not complete for this DTC bit not supported. Set to 0b.
5..6	• DTCStorageState – Storage state of the error path: • 00b = “No DTC detected at time of request” • 01b = “Error path is stored in the FCM but currently not defective” • 10b = “DTC maturing – intermittent at time of request” • 11b = “Error path is stored in the FCM and currently defective”
7	• DTCWarningLampStatus, indicates whether a warning lamp is controlled by the path entered • 0b = “Error path is not controlling a warning lamp” • 1b = “Error path is controlling a warning lamp”

PARAMETER: NumberOfFaultDetections		
Size (Byte)	**Description**	**Value**
1	If a fault remains current in each new ignition cycle, NumberOfFaultDetections gets increased by one until the fault is not deleted. (max value is FFh)	xxh

Remark: The fault path is indicated by the stored DTC. When saving a new fault, a check is first conducted in order to establish whether the same DTC has already been saved at any point in the fault memory. If this is the case, the DTC is saved at the same position. Otherwise, it is saved at the next free position in the fault memory.

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6.4.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	17h
1	DTC – High Byte	xxh
2	DTC – Middle Byte	xxh
3	DTC – Low Byte	xxh

6.4.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	57h
1	NumberOfDTC	01h
2	DTC – High Byte	xxh
3	DTC – Middle Byte	xxh
4	DTC – Low Byte	xxh
5	StatusOfDTC	xxh
6	NumberOfFaultDetections	xxh
7	Vehicle Speed (1LSB = 1km/h, FFh = Invalid)	xxh
8	ECU power supply (1LSB = 80mV, FFh = Invalid)	xxh

6.4.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	17h
2	**Response Code** Sub Function Not Supported / Invalid Format BusyRepeatRequest RequestCorrectlyReceived/ResponsePending	 12h 21h 78h

Remark: If there is no saved fault corresponding to the requested DTC, the ECU is sending a negative response with the parameter 12h.

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6.5 ReadDTCByStatus

6.5.1 Purpose

The service ReadDTCByStatus is used to read all stored DTCs from the failure memory.

StatusOfDTC	
Position (Bit)	**Description**
0..3	DTCFaultSymptom • 0000b = “No fault symptom available for this DTC” • 0001b = “Above maximum threshold” • 0010b = “Below minimum threshold” • 0100b = “No signal” • 1000b = “Invalid signal”
4	• 0b = Test completed / not complete for this DTC bit not supported. Set to 0b.
5..6	DTCStorageState – Storage state of the error path: • 00b = “No DTC detected at time of request” • 01b = “Error path is stored in the FCM but currently not defective” • 10b = “DTC maturing – intermittent at time of request” • 11b = “Error path is stored in the FCM and currently defective”
7	DTCWarningLampStatus, indicates whether a warning lamp is controlled by the path entered • 0b = “Error path is not controlling a warning lamp” • 1b = “Error path is controlling a warning lamp”

Remark: The entire fault memory of the ECU can accommodate up to 10 different errors simultaneously.

Remark: Old DTCs will be deleted after 200 ignition cycle. The Ignition cycle of the fault and current ignition cycle will be compared. If the difference is greater than or equal to 200, then the fault is deleted from the EEPROM.

6.5.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	18h
1	StatusOfDTC	00h
2	GroupOfDTC – High Byte	FFh
3	GroupOfDTC – Low Byte	00h

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6.5.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	58h
1	NumberOfDTC	xxh
2..4	DTC	xxxxxxh
5	StatusOfDTC	xxh
…	DTC	xxxxxxh
…	StatusOfDTC	xxh

Remark: According to the NumberOfDTC more than one DTC can be sent.

Remark: For every DTC 4 bytes are sent.

Remark: If no DTC is saved the response ends at byte 1 with NumberOfDTC at the value 00h.

6.5.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Content**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	18h
2	**Response Code** Sub Function Not Supported / Invalid Format BusyRepeatRequest	 12h 21h

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6.6 ReadECUIdentification

6.6.1 Purpose

The service ReadECUIdentification is used to read identification information from the ECU.
The parameter IdentificationOption is used to indicate which information has to be read.

PARAMETER: IdentificationOption		
Size (Byte)	**Description**	**Value**
1	ReportECUIdentificationTable	80h

6.6.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	1Ah
1	IdentificationOption	80h

6.6.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	5Ah
1	Local Identifier	80h
2..4	Vehicle Name (ASCII, 3 bytes) → A01 : 0x41 0x30 0x31	xxxxxxh
5..7	System Name (ASCII, 3 bytes) → ABS : 0x41 0x42 0x53	xxxxxxh
8..10	BBNumber (BCD, 3 Bytes) → BB90543: 0x09 0x05 0x43	xxxxxxh
11..14	S/W Version (BCD, 4 bytes) → V01.02.01.01 : 0x01 0x02 0x01 0x01	xxxxxxxxh
15..19	Supplier Specific Part Number (BCD, 5 bytes) → 0265805057 : 0x02 0x65 0x80 0x50 0x57	xx…xxh
20..32	Customer Specific Part Number (ASCII) 13 bytes 3550100-A01 : 0x33 0x35 0x35 0x30 0x31 0x30 0x30 0x2D 0x41 0x30 0x31 0x20 0x20	xx…xxh

Remark: Byte 15... 32 will be read from EEPROM.
Remark: If the customer part number length is less than the allocated bytes, then the remaining bytes will be updated with the value 0x20(ASCII value for SPACE).

6.6.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Content**	**Value**
0	Negative response Identifier	7Fh
1	Request Service Identifier	1Ah
2	**Response Code** General Reject SubFunctionNotSupported/InvalidFormat ResponsePending	 10h 12h 78h

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Remark: If the reading of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has finished.

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6.7 ReadDataByLocalIdentifier

6.7.1 Purpose

This service allows the tester to request current data from the ECU using specific data IDs.

PARAMETER: RecordLocalidentifier			
Request Data byte (Size)	**Description**	**Response Databyte (Size)**	**Value (Record LocalIdentifier)**
1	Supported PID for 01 – 20 ABS → 7F 81 80 10	4	00h
1	Filling-in status of the hydraulic block Filling-in status: • 00h : Filling-in not completed • AAh : Filling-in completed and ok • EEh : Filling-in completed and not ok	1	02h
1	Test of End-Of-Line End of Line Test status: • 00h : End of Line Test not completed • AAh : End of Line Test completed and ok • EEh : End of Line Test completed and not ok	1	03h
1	Wheel Speed FL Range 0…255 km/h 1LSB = 0.0563km/h	2	04h
1	Wheel Speed FR Range 0…255 km/h 1LSB = 0.0563 km/h	2	05h
1	Wheel Speed RL Range 0…255 km/h 1LSB = 0.0563 km/h	2	06h
1	Wheel Speed RR Range 0…255 km/h 1LSB = 0.0563 km/h	2	07h
1	Vehicle Speed Range 0…255 km/h 1LSB = 0.0563 km/h	2	08h
1	Valve Status Byte 1: Bit 0: EVFL - Inlet valve front left - 1h ON / 0h OFF Bit 1: AVFL - Outlet valve front left - 1h ON / 0h OFF Bit 2: EVFR - Inlet valve front right - 1h ON / 0h OFF Bit 3: AVFR - Outlet valve front right - 1h ON / 0h OFF Bit 4: EVRL - Inlet valve rear left - 1h ON / 0h OFF Bit 5: AVRL - Outlet valve rear left - 1h ON / 0h OFF Bit 6: EVRR - Inlet valve rear right - 1h ON / 0h OFF Bit 7: AVRR - Outlet valve rear right - 1h ON / 0h OFF	1	09h
1	Pump Motor Status FFh ON / 00h OFF	1	10h
1	BLS Status FFh ON / 00h OFF	1	11h
1	Battery Voltage Range 8…19.89V 1LSB = 0.080 V/bit	1	1Ch

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6.9 InputOutputControlByLocalIdentifier

This request is used to actuate a single control. The parameter ControlOption is used to indicate whether the control has to be temporary, permanent or whether it has to be stopped. When using ControlOption "Start temporary control" the control runs for 2s. When using ControlOption "Start permanent control" the control runs until it is stopped by sending the same request with ControlOption set to "Stop control".

Supported LIDs for Single Output Control (InputOutputLocalIdentifier)

PARAMETER: InputOutputControlByLocalIdentifier				
Size (Byte)	**Description**	**Value**	**Temp.**	**Perm.**
1	Single control - Motor (Pump)	01h	Yes	No
1	Single control - Front left valve (inlet)	02h	Yes	No
1	Single control - Front right valve (inlet)	03h	Yes	No
1	Single control - Rear left valve (inlet)	04h	Yes	No
1	Single control - Rear right valve (inlet)	05h	Yes	No
1	Single control - Front left valve (outlet)	06h	Yes	No
1	Single control - Front right valve (outlet)	07h	Yes	No
1	Single control - Rear left valve (outlet)	08h	Yes	No
1	Single control - Rear right valve (outlet)	09h	Yes	No
1	Single control - Speed limit disable (for $V_{Veh} \geq 10$km/h)	1Eh	No	Yes
1	Single control - Disable valve relay shutdown	1Fh	No	Yes

6.9.1 Request

TESTER → ECU: [SERVICE] - Request		
Byte	**Contents**	**Value**
0	ServiceIdentifier	30h
1	InputOutputLocalIdentifier	xxh
2	**InputOutputControlParameter** Start temporary control 01h Stop control 02h Status interrogation 03h Start permanent control 04h	xxh

Two ways of control can be distinguished:

Temporary: The actuator will be switched on for 2 s.
Permanent: The actuator will be switched on. It can be switched off (stopped) by sending a "Emergency Stop Request" (command 02H).

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6.9.2 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	70h
1	InputOutputLocalIdentifier	xxh
2	**Status** Started 01h Control completed 02h Control In Progress 03h Stopped 04h Failure Or Not Started 05h	xxh

Remarks

- If the tester is trying to start a **new control while another actuation is still being executed**, the ECU responds with the Positive response – Failure Or Not started (05h).This is not applicable for temporary control while Permanent control is in progress and vice versa. If the tester is trying to start a **new control while the same actuation is still being executed**, the ECU responds with the Positive response- Control in Progress(03h).
- **LIDs $1E and $1F should be used with care**, these LIDs will disable parts of the safety software. Therefore, the ECU may be damaged or the vehicle may not be safe to drive (e.g. if the speed limit during diagnostics is disabled and then the valves are activated while driving the car).

If the valves are activated for long time, A valve overheat protection functionality is implemented. When Valves are actuated for more than a specified time then actuation will be disabled .When Valves cool down actuation will be activated again. This is done through up count and down count mechanism. Overheat protection for valves avoid failures in the field.

Remark: After a start permanent control request in combination with Input Output Control By Local Identifier values 1Eh and 1Fh the ECU will respond to a status interrogation request in combination with Input Output By Local Identifier values 1Eh and 1Fh with a positive response including the ControlStatus 03h (control In Progress).

6.9.3 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	30h
2	**Response Code** Sub Function Not Supported / Invalid Format Conditions Not Correct / Request Sequence Error	 12h 22h

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6.10 StartRoutineByLocalIdentifier

6.10.1 Purpose

The service StartRoutineByLocalIdentifier is used to control actuators connected to the ECU. Depending on the parameter StartRoutineLocalIdentifier a specific function is started (01h to 06h). The following values for the parameter StartRoutineLocalIdentifier are defined:

Size (Byte)	Description	Value
1	GetSupportedLID	00h
1	DelayedActuatorControl	01h
1	Evac&Fill Routine	02h
1	RepairBleed Control	03h
1	ActuatorTest	04h
1	DynamicTest	05h
1	Test of the wheel speed sensors	06h

PARAMETER: ControlOption		
Size (Byte)	**Description**	**Value**
1	Start Routine	01h
1	Stop Routine	02h
1	Request routine results	03h

6.10.2 Requests

Remark: If the valves are activated for too long the ECU can be damaged. The maximum activation time of the valves must not exceed 30 seconds at an ambient temperature of 30°C. Note that Bosch has changed the overheat protection strategy for system 9 (outlet valves no longer toggling after 1s). Using this service repeatedly can damage the valves or pump. Please keep this information in mind when using this service.

Remark: If the ControlOption is set to “Stop control” the request must only have 3 bytes: Service Identifier, StartRoutineLocalIdentifier, ControlOption.

Remark: For control actions with an executing time exceeding $P2_{CAN\text{-}ECU}$ the ECU normally responds positive with ControlStatus = Control in progress within $P2_{CAN\text{-}ECU}$. During executing time the tester has to ask for the control result by using the ControlOption = Status interrogation. The ECU will either send a response with the ControlStatus = Control in progress or with the ControlStatus = Control complete and in addition if applicable with a result. The service requests TesterPresent and EmergencyStop will as well be positively responded.

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6.10.3 Emergency Stop

At any time it is possible to interrupt the control of the output which is being tested.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	xxh
2	ControlOption = Stop Control	02h

6.10.4 StartRoutineLocalIdentifier = GetSupportedLID

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	00h

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6.10.5 StartRoutineLocalIdentifier = DelayedActuatorControl

This request is used to control multiple drivers simultaneously. For each control two states are defined: state 1 and state 2. When the request was received the selected controls are set in state 1. After the activation time they are set to state 2.

In comparison to the Actuator Test this service allows the tester to actuate one or more valves for a specified time. This feature allows the tester to run the customer specified End Of Line routine sequence instead of Pre-defined Periodic Actuator Control sequence.

PARAMETER: ValveStates	
Position (Bit)	**Description**
0	Inlet front left
1	Outlet front left
2	Inlet front right
3	Outlet front right
4	Inlet rear left
5	Outlet rear left
6	Inlet rear right
7	Outlet rear right

PARAMETER: DriverStates	
Position (Bit)	**Description**
0	Unused
1	Unused
2	Unused
3	Unused
4	Unused
5	Unused
6	Pump
7	Unused

Remark: For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.

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TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	01h
2	ControlOption	01h
3	ValveStates (State 1)	xxh
4	DriverStates (State 1)	xxh
5	Activation time (1LSB = 20ms)	xxh
6	ValveStates (State 2)	xxh
7	DriverStates (State 2)	xxh

Remark: A "Failure" response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.

Remark: A control that has been started by a delay actuator control request can be stopped by a request with the ControlOption set to "Stop control" (Emergency Stop, 31h 01h 02h).

Remark: At each end of this service the tester has to send the following request to turn off all controls: 31h 01h 01h 00h 00h 00h 00h 00h.

6.10.6 StartRoutineLocalIdentifier = EvacAndFill

This request is used to fill and drain the hydraulic unit.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	02h
2	ControlOption	01h
3	Evac and fill time	xxh
4	Time until start of pump motor	xxh

Remark: Evac &Fill time has to be between 2 s and 150 s. If the requested time > 150 seconds, the ECU uses the maximum allowed value(150 s). If the requsted time<2seconds,the Routine will not run. Only pump run will happen based on the byte4(parameter 'Time until start of pump motor'). The duration time selection only supports multiple of 2 (as one EvacAndFill cycle takes 2 seconds). Odd numbers are cut down to the next lower even value (i.e. if the routine is requested to run 15 s, it will actually complete in 14 s).After evac and filling time, the pump will run for 2 additional seconds,if the delay time in byte4 is not FFh.If total Evac&Fill time <' Time until start of pump motor' ,then the ECU responds Negatively with NRC 12h.

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6.10.7 StartRoutineLocalIdentifier = RepairBleeding

This request is used to fill and drain the hydraulic unit.

TESTER → ECU: [SERVICE] – Request			
Byte	**Contents**	**Value**	**Comment**
0	Service Identifier	31h	
1	StartRoutineLocalIdentifier	03h	
2	ControlOption	01h	
3	Repair bleeding phase (1..4)	xxh	Phase 1..4
4	Repetition counter(3..5)	xxh	No. of loops: 3..5

Remark:
Repair bleeding phase
The repair bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not assured, the pump might be damaged due to dry run.
Repetition counter
This parameter defines the number of inner loop repetitions in phase 3and 4. This parameter is always required, but only valid for Phase 3 and 4, for other phases, it is ignored.

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6.10.8 Parameter descriptions for Actuator Test and Dynamic Test

Following parameter descriptions are valid for the ActuatorTest and for the Dynamic Test

PARAMETER: ResultSelection		
Size (Byte)	**Description**	**Value**
1	Without result	00h
1	With result	FFh

PARAMETER: ActuationValue		
Size (Byte)	**Description**	**Value**
2	This parameter specifies the amount of actuation. Resolution: 1 bit = 1%	xxxxh

PARAMETER: ActuatorNumber		
Size (Byte)	**Description**	**Value**
2	No actuation*	0000h
2	Inlet valve front left	0030h
2	Outlet valve front left	0032h
2	Inlet valve front right	0034h
2	Outlet valve front right	0036h
2	Inlet valve rear right	0038h
2	Outlet valve rear right	003Ah
2	Inlet valve rear left	003Ch
2	Outlet valve rear left	003Eh
2	Pump motor relay	0022h
2	ABS Safety Lamp	0002h
2	EBD Safety Lamp	000Eh

Remark: * Use if less than four actuators are to be actuated by an actuator test or dynamic test request.

PARAMETER: ActuationTime		
Size (Byte)	**Description**	**Value**
2	This parameter specifies the time between the first and the second actuation. Resolution: 1 bit = 10 ms	Xxxxh

PARAMETER: WaitTime		
Size (Byte)	**Description**	**Value**
2	This parameter specifies the waiting time. Resolution: 1 bit = 10 ms	Xxxxh

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6.10.9 StartRoutineLocalIdentifier = ActuatorTest

This request is used to start the ActuatorTest.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	04h
2	ControlOption	01h
3	ResultSelection	Xxh
4..5	ActuationValue 1	Xxxxh
6..7	ActuatorNumber 1	Xxxxh
8..9	ActuationValue 2	Xxxxh
10..11	ActuatorNumber 2	Xxxxh
12..13	ActuationTime (1LSB = 10ms)	Xxxxh
14..15	ActuationValue 3	Xxxxh
16..17	ActuationNumber 3	Xxxxh
18..19	ActuationValue 4	Xxxxh
20..21	ActuatorNumber 4	Xxxxh

Remark: The pump motor relay must not be actuated for times shorter than 20ms. Shorter actuations could damage the motor relay due to very high current during the starting phase of the motor.

Remark: Bosch strongly recommends only using an ActuationValue of 0% and 100%.

Remark: A negative response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.

Remark: If the valve relay is out of order, the routine will be aborted.

Example:

- Start actuator test **without result** selection

Request	Response	Comment
31 04 01 **00** Xx Xx ...	71 04 01	-> Start actuation, Routine started by the tester
31 04 03	71 04 03	-> Status interrogation, control still in progress
.	.	
.	.	
31 04 03	71 04 02	-> Status interrogation, control complete, no result bytes sent

- Start actuator test **with result** selection

Request	Response	Comment
31 04 01 **FF** Xx Xx ...	71 04 01	-> Start actuation, routine started by the tester
31 04 03	71 04 03	-> Status interrogation, control still in progress
.	.	
31 04 03	71 04 02 Xx Xx ...	-> Status interrogation, control complete, result bytes sent

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6.10.10 StartRoutineLocalIdentifier = DynamicTest

This request is used to start the Dynamic Test.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	05h
2	ControlOption	01h
3..4	ActuationValue 1	Xxxxh
5..6	ActuatorNumber 1	Xxxxh
7..8	ActuationValue 2	Xxxxh
9..10	ActuatorNumber 2	Xxxxh
11..12	ActuationTime (1LSB = 10ms)	Xxxxh
13..14	ActuationValue 3	Xxxxh
15..16	ActuationNumber 3	Xxxxh
17..18	ActuationValue 4	Xxxxh
19..20	ActuatorNumber 4	Xxxxh
21..22	WaitTime (1LSB = 10ms)	Xxxxh

6.10.11 StartRoutineLocalIdentifier = Test of the wheel speed sensors

This request is used to start the wheel speed sensor test.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	06h
2	ControlOption	01h
3	Duration (1LSB = 100ms)	xxh

6.10.11.1 RequestRoutineResult

The Request Routine Results by Local Identifier gets the optional results of routines previously activated with the Start Routine.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	31h
1	StartRoutineLocalIdentifier	xxh
2	ControlOption = Routine result	03h

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6.10.12 Positive Response

PARAMETER: Control/RoutineStatus		
Size (Byte)	**Description**	**Value**
1	Control/Routine Started successfully by the tester	01h
1	Control/Routine completed	02h
1	Control/Routine in progress	03h
1	Control/Routine stopped	04h
1	Control/Routine failure or Routine not run	05h

6.10.12.1 Control Option = Start Routine and Stop Routine

For requests with Control option Start Routine (01h) and Stop Routine (02h), the positive response is returned in the following format.

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	xxh
2	ControlStatus • Routine successfully Started 01h • Routine in Progress 03h • Routine Stopped 04h • Routine Failure or Routine Not Run 05h	xxh

Remarks:

- If the Tester send the request to Stop the Routine after routine is completed ,Positive Response(Control status =05h) will be returned
- If the Tester send the request to start the routine while another routine is in progress, Positive Response(Control status =05h) will be returned
- If the Tester send the request to start the routine while any temporary control in IOControlByLID service(SID 30h) is in progress, Positive Response(Control status =05h) will be returned.
- If the Tester send the request to start the routine while the same routine is in progress, Response(Control status =03h) will be returned.

6.10.12.2 StartRoutineLocalIdentifier = GetSupportedLID

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	00h
2	Data A describes the support of LID 01h ... 08h (MSB)	FCh
3	Data B describes the support of LID 09h ... 10h	00h
4	Data C describes the support of LID 11h ... 18h	00h
5	Data D describes the support of LID 19h ... 20h (LSB)	00h

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Remark: Supported PID (00h) gives the information about the supported local identifier from 01h – 20h (Bit 31 - PID 01h, Bit 30 – PID 02h … Bit 0 – PID 20h).

6.10.12.3 StartRoutineLocalIdentifier = Delayed Actuator, EvacAndFill and RepairBleeding

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier (01h ,02h and 03h)	Xxh
2	ControlStatus	Xxh

6.10.12.4 StartRoutineLocalIdentifier = Actuator Test, response without result

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	04h
2	ControlStatus	Xxh

6.10.12.5 StartRoutineLocalIdentifier = Actuator Test, response with result

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	04h
2	ControlStatus	Xxh
3..4	Result, actuator 1	Xxxxh
5..6	Result, actuator 2	Xxxxh
7..8	Result, actuator 3	Xxxxh
9..10	Result, actuator 4	Xxxxh

Remark: Result is a signal which informs about the former actuation of the corresponding actuator.
Remark: Bytes 3…10 are only sent if the response is “RoutineStatus = Routine completed”.

6.10.12.6 StartRoutineLocalIdentifier = Dynamic Test

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	05h
2	ControlStatus	Xxh
3..4	Result, delta wheel speed front left (1LSB = (1/64)m/s)	Xxxxh
5..6	Result, delta wheel speed front right (1LSB = (1/64)m/s)	Xxxxh
7..8	Result, delta wheel speed rear left (1LSB = (1/64)m/s)	Xxxxh
9..10	Result, delta wheel speed rear right (1LSB = (1/64)m/s)	Xxxxh

Remark: Result is a signed word information with a resolution of 1 LSB = 1/64 m/s = 0,05625 km/h and a range of -100 m/s … x … 100 m/s or -360 km/h … x … 360 km/h.
Remark: Bytes 3…10 are only sent if the response is “RoutineStatus = Routine completed”.

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6.10.12.7 StartRoutineLocalIdentifier = Test of the wheel speed sensors

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	71h
1	StartRoutineLocalIdentifier	06h
2	RoutineStatus	xxh
3..4	Maximum speed of the front left wheel (1LSB = 0,01km/h)	xxxxh
5..6	Minimum speed of the front left wheel (1LSB = 0,01km/h)	xxxxh
7..8	Maximum speed of the front right wheel (1LSB = 0,01km/h)	xxxxh
9..10	Minimum speed of the front right wheel (1LSB = 0,01km/h)	xxxxh
11..12	Maximum speed of the rear left wheel (1LSB = 0,01km/h)	xxxxh
13..14	Minimum speed of the rear left wheel (1LSB = 0,01km/h)	xxxxh
15..16	Maximum speed of the rear right wheel (1LSB = 0,01km/h)	xxxxh
17..18	Minimum speed of the rear right wheel (1LSB = 0,01km/h)	xxxxh

Remark: Bytes 3...18 are only sent if the response is “RoutineStatus = Routine completed”.

Remark: The test should be done during at least one complete wheel rotation.
Remark: If the test fails the minimum and maximum speed of a wheel speed sensor are set to 7FFFh.

Remark: For routines with an executing time exceeding $P2_{CAN\text{-}ECU}$ the ECU normally responds positive with RoutineStatus = Routine in progress within $P2_{CAN\text{-}ECU}$. During executing time the tester has to ask for the routine result by using the ControlOption = Request Routine Results. The ECU will either send a response with the RoutineStatus = Routine in progress or with the RoutineStatus = Routine completed and in addition if applicable with a result. The service request TesterPresent will as well be positively responded.

6.10.13 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	RequestServiceIdentifier	31h
2	**ResponseCode** SubFunctionNotSupported/InvalidFormat ConditionsNotCorrect/RequestSeqError	 12h 22h

Remarks

- At the end of each command, the elements which were tested have to be replaced in their OFF states.

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6.11 WriteDataByLocalIdentifier

6.11.1 Purpose

The service WriteDataByLocalIdentifier is used to write information into the ECU. The parameter RecordLocalIdentifier is used to select which information has to be written.

PARAMETER: RecordLocalIdentifier		
Size (Byte)	**Description**	**Value**
1	Filling-in status of the hydraulic block	02h
1	Status of Test of end-of-line	03h

6.11.2 Request

6.11.2.1 RecordLocalIdentifier = Filling-in status of the hydraulic block

This request is used to write a byte which indicates the filling-in status of the hydraulic block.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	3Bh
1	RecordLocalIdentifier	02h
2	Filling-in status: • 00h : Filling-in not completed • AAh : Filling-in completed and ok • EEh : Filling-in completed and not ok	xxh

Remark: Bosch delivery state is FFh.

6.11.2.2 RecordLocalIdentifier = Status of Test of End-Of-Line

This request is used to write a byte which indicates the status if the end-of-line test.

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	3Bh
1	RecordLocalIdentifier	03h
2	End-of-line test status: • 00h : End-of-line test not completed • AAh : End-of-line test completed and ok • EEh : End-of-line test completed and not ok	xxh

Remark: Bosch delivery state is FFh.

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6.11.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	7Bh
1	RecordLocalIdentifier	xxh

6.11.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	3Bh
2	**Response Code** General Reject Sub Function Not Supported / Invalid Format Conditions Not Correct / Request SequenceError Response Pending Security Access Denied	 10h 12h 22h 78h 33h

Remark: If the reading of the EEPROM needs more than $P2_{CAN\text{-}ECU}$, the ECU will send a negative response 78h (RequestCorrectlyReceived-ResponsePending) until it has finished.

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6.12 TesterPresent

6.12.1 Purpose

The service TesterPresent is used to indicate the ECU that a tester is connected and diagnostic services and/or communication have to remain active. Sending the request keeps the ECU in a diagnostic session other than the StandardDiagMode.

PARAMETER: ResponseRequired		
Size (Byte)	**Description**	**Value**
1	ResponseRequired	01h
1	NoResponseRequired	00h

6.12.2 Request

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	3Eh
1	ResponseRequired	01h

TESTER → ECU: [SERVICE] – Request		
Byte	**Contents**	**Value**
0	Service Identifier	3Eh
1	NoResponseRequired	00h

6.12.3 Positive Response

ECU → TESTER: [SERVICE] - Positive Response		
Byte	**Contents**	**Value**
0	Service Positive Response Identifier	7Eh
1	ResponseRequired	01h

6.12.4 Negative Response

ECU → TESTER: [SERVICE] - Negative Response		
Byte	**Contents**	**Value**
0	Negative Response Identifier	7Fh
1	Request Service Identifier	3Eh
2	SubFunctionNotSupported/InvalidFormat	12h

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7 List of DTCs

7.1 DTC List

DTC	DTC (Hex)	Fault Symptoms	DTC Description
C1900	0x590004	0100	ECU Voltage supply: high voltage
C1901	0x590104	0100	ECU Voltage supply: low voltage
C1000	0x500004	0100	ECU error (hardware, Micro-controller error)
C1010	0x501008	1000	ECU error (general software problem)
C006B	0x406B06	0110	ABS implausible control (control time too long)
C0031	0x403108	1000	Wheel-speed sensor, front left: (signal failure) out of range, lost, noise, Intermittent
C0032	0x403200	0000	Wheel-speed sensor, front left: open
C00A0	0x40A000	0000	Wheel-speed sensor, front left: short to GND
C00A1	0x40A100	0000	Wheel-speed sensor, front left: short to UBATT
C00A9	0x40A900	0000	Wheel-speed sensor, front left: undef (no exact failure reason detectable)
C0034	0x403408	1000	Wheel-speed sensor, front right: (signal failure) out of range, lost, noise, Intermittent
C0035	0x403500	0000	Wheel-speed sensor, front right: open
C00A2	0x40A200	0000	Wheel-speed sensor, front right: short to GND
C00A3	0x40A300	0000	Wheel-speed sensor, front right: short to UBATT
C00AA	0x40AA00	0000	Wheel-speed sensor, front right: undef (no exact failure reason detectable)
C0037	0x403708	1000	Wheel-speed sensor, rear left: (signal failure)out of range, lost, noise, Intermittent
C0038	0x403800	0000	Wheel-speed sensor, rear left: open
C00A4	0x40A400	0000	Wheel-speed sensor, rear left: short to GND
C00A5	0x40A500	0000	Wheel-speed sensor, rear left: short to UBATT
C00AB	0x40AB00	0000	Wheel-speed sensor, rear left: line failure (no exact failure reason detectable)
C003A	0x403A08	1000	Wheel-speed sensor, rear right: (signal failure)out of range, lost, noise, Intermittent
C003B	0x403B00	0000	Wheel-speed sensor, rear right: open
C00A6	0x40A600	0000	Wheel-speed sensor, rear right: short to GND
C00A7	0x40A700	0000	Wheel-speed sensor, rear right: short to UBATT
C00AC	0x40AC00	0000	Wheel-speed sensor, rear right: line failure (no exact failure reason detectable)
C1099	0x509904	0100	Wheel-speed sensor generic error(sensor swap, multiple sensors error）
U0005	0xC00500	0000	CAN Bus overvoltage
U0007	0xC00700	0000	CAN Bus under voltage
C1001	0x500104	0100	CAN hardware error
U0001	0xC00104	0100	CAN Bus Off error
U1001	0xD10104	0100	CAN Error Passive
C0010	0x401004	0100	valve fault, inlet valve FL

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DTC	DTC (Hex)	Fault Symptoms	DTC Description
C0011	0x401104	0100	valve fault, outlet valve FL
C0014	0x401404	0100	valve fault, inlet valve FR
C0015	0x401504	0100	valve fault, outlet valve FR
C0018	0x401804	0100	valve fault, inlet valve RL
C0019	0x401904	0100	valve fault, outlet valve RL
C001C	0x401C04	0100	valve fault, inlet valve RR
C001D	0x401D04	0100	valve fault, outlet valve RR
C1095	0x509504	0100	Valve relay error
C0020	0x402004	0100	Return pump fault
C0072	0x407208	1000	General Valve fault (overheat protection, signal invalid, hardware fault)

NOTE:

Possible Fault Symptoms corresponds to

1) 0100 – No Signal
2) 1000 – Invalid Signal
3) 0000 – No fault symptom available for this DTC

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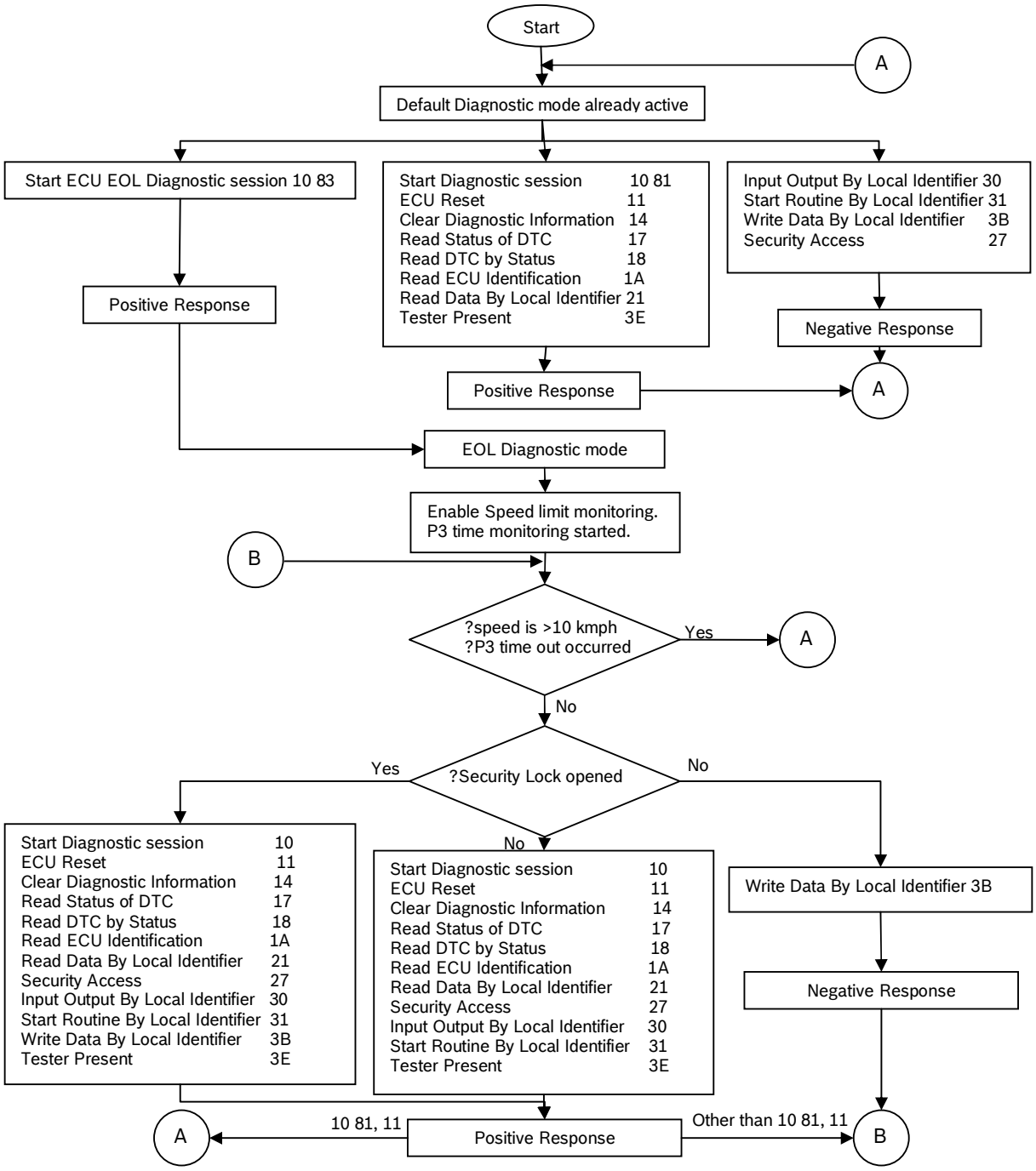
8 Diagnostic Flow Chart



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9 Appendix – A: Actuator Table

Value	Description	Resolution/Comment
0000h	No actuation*	0x0000
0022h	Single control - Motor (Pump)	0xFFFF – ON 0x0000 - OFF
0030h	Single control - Front left valve (inlet)	0xFFFF – ON 0x0000 – OFF
0032h	Single control - Front left valve (outlet)	0xFFFF – ON 0x0000 – OFF
0034h	Single control - Front right valve (inlet)	0xFFFF – ON 0x0000 – OFF
0036h	Single control - Front right valve (outlet)	0xFFFF – ON 0x0000 – OFF
0038h	Single control - Rear right valve (inlet)	0xFFFF – ON 0x0000 – OFF
003Ah	Single control - Rear right valve (outlet)	0xFFFF – ON 0x0000 - OFF
003Ch	Single control - Rear left valve (inlet)	0xFFFF – ON 0x0000 – OFF
003Eh	Single control - Rear left valve (outlet)	0xFFFF – ON 0x0000 – OFF
0002h	Single control – ABS Safety Lamp	0xFFFF – ON 0x0000 – OFF
000Eh	Single control – EBD Safety Lamp	0xFFFF – ON 0x0000 - OFF

Remark: * Use if less than four actuators are to be actuated by an actuator test or dynamic test request.

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